Curcumin: an orally bioavailable blocker of TNF and other pro-inflammatory biomarkers

TNFs are major mediators of inflammation and inflammation-related diseases, hence, the United States Food and Drug Administration (FDA) has approved the use of blockers of the cytokine, TNF-α, for the treatment of osteoarthritis, inflammatory bowel disease, psoriasis and ankylosis. These drugs include the chimeric TNF antibody (infliximab), humanized TNF-α antibody (Humira) and soluble TNF receptor-II (Enbrel) and are associated with a total cumulative market value of more than $20 billion a year. As well as being expensive ($15 000–20 000 per person per year), these drugs have to be injected and have enough adverse effects to be given a black label warning by the FDA. In the current report, we describe an alternative, curcumin (diferuloylmethane), a component of turmeric (Curcuma longa) that is very inexpensive, orally bioavailable and highly safe in humans, yet can block TNF-α action and production in in vitro models, in animal models and in humans. In addition, we provide evidence for curcumin's activities against all of the diseases for which TNF blockers are currently being used. Mechanisms by which curcumin inhibits the production and the cell signalling pathways activated by this cytokine are also discussed. With health-care costs and safety being major issues today, this golden spice may help provide the solution. Linked Articles This article is part of a themed section on Emerging Therapeutic Aspects in Oncology. To view the other articles in this section visit http://dx.doi.org/10.1111/bph.2013.169.issue-8